Syllabus Details

Syllabus ID:	12400
Syllabus Name:	**Internet of Things_Internet vạn vật**
Course Name English:	**Internet of Things**
Subject Code:	**IOT102**
Learning-Teaching Method:	Social Constructivism, Inquiry-Based Learning, In-class lecture, Active Learning, Project-based learning
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 22.5h contact hours + 22.5h online hours + 4.5h final project presentation + 1h final exam + 99.5h self-study
Pre-Requisite:	
Description:	This is a 3-credit course. This course has two parts: online and offline. The content includes basic concepts and applications of IoT and practical exercises on learning KIT. Students are taught how to learn online and practice some parts at home. Q&A sessions, Guidance for important issues, and performance assessments will be conducted in the classroom.
StudentTasks:	- Students must attend at least 80% of contact sessions in order to be accepted to the final examination. - Student is responsible to learn all VIDEOs (theory) online given by instructor on Syllabus at home. - Student is responsible to do all LABs given by instructor on Syllabus at home. - Constantly follow https://flm.fpt.edu.vn/ for up-to-date course information.
Tools:	
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	1363/QĐ-ĐHFPT dated 12/06/2024
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	12/6/2024

7 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	Course from Edx						True	False	True	https://learning.edx.org/course/course-v1:CurtinX+IOT1x+2T2018/home
2	Course from Edx						True	False	True	https://learning.edx.org/course/course-v1:CurtinX+IOT2x+2T2018/home
3	Free Hardware and IoT Ebooks_ OReilly Media						False	False	False	soft copy
4	Introduction to Arduino: A piece of cake! by Alan G. Smith, September 30, 2011						False	False	True	http://www.introtoarduino.com
5	Arduino Documents						False	False	True	https://www.arduino.cc/
6	Instructables.com - 20 Unbelievable Arduino Projects						False	False	False	soft copy

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
7	IOT102 guide documents (Project guide, Project topics, Rubrics, KITs (basic and advanced)...)						True	False	False	soft copy

13 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Understand the concept of IoT and applications.
2	CLO2	Understand the concept of things in IoT.
3	CLO3	Understand the Networking in IoT.
4	CLO4	Understand the Securing in IoT.
5	CLO5	Select and calculate the electronics components, IoT Programming.
6	CLO6	Program IN/OUT Digital with Arduino UNO.
7	CLO7	Program Analog Input/Output with Arduino UNO.
8	CLO8	Program Sensors/Actuators with Arduino UNO.
9	CLO9	Program Communication with Arduino UNO.
10	CLO10	Concept planning, Draw and Install simple IoT circuits
11	CLO11	Understand the Arduino UNO and Atmega328 microcontroller architecture.
12	CLO12	Know how to use the Simulation Software.
13	CLO13	Understand the interrupts on Arduino UNO.

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Introduction of IoT course What in the world is the Internet of Things?	Offline	CLO01		Slide Video GUIDE Of Create an Account_Explore Courses_Quiz on edX.pdf	IOT102	Borrow theLearning KIT at the library. Enroll course: https://learning.edx.org/course/course-v1:CurtinX+IOT1x+2T2018/home	
2	Arduino (ATmega328p) Architecture	Offline	CLO01		Slide Video GUIDE Of Create an Account_Explore Courses_Quiz on edX.pdf		Borrow theLearning KIT at the library. Enroll course: https://learning.edx.org/course/course-v1:CurtinX+IOT1x+2T2018/home	
3	Lab: - Getting Started with Arduino and Genuino UNO - Fritzing_building_circuit	Offline	CLO10, CLO12		Tutorial Learning KIT		Discuss with your instructor the lessons learned online and the new lesson.	
4	The 'things' of the Internet of Things?	Online	CLO02		Slide Video		Enroll course: https://learning.edx.org/course/course-v1:CurtinX+IOT1x+2T2018/home Follow the lessons online	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
5	Networking IoT	Online	CLO03		Slide Video		Enroll course: https://learning.edx.org/course/course-v1:CurtinX+IOT1x+2T2018/home Follow the lessons online	
6	Networking IoT (cnt)	Online	CLO03		Slide Video		Enroll course: https://learning.edx.org/course/course-v1:CurtinX+IOT1x+2T2018/home Follow the lessons online	
7	Lab: - Tinkercad arduino simulator tutorial	Offline	CLO10, CLO12		Tutorial Learning KIT		Discuss with your instructor the lessons learned online and the new lesson.	
8	Lab: - Blink an LED - Digital Write - Analog Input - Analog Read	Offline	CLO06, CLO07		Tutorial Learning KIT		Discuss with your instructor the lessons learned online and the new lesson.	
9	Lab: - Fading - Analog Write Project Topic Introduction	Offline	CLO06, CLO07		Tutorial Learning KIT		Discuss with your instructor the lessons learned online and the new lesson.	
10	Programming IoT	Online	CLO01		Slide Video		Enroll course: https://learning.edx.org/course/course-v1:CurtinX+IOT1x+2T2018/home Follow the lessons online	
11	Programming IoT (cnt)	Online	CLO01		Slide Video		Enroll course: https://learning.edx.org/course/course-v1:CurtinX+IOT1x+2T2018/home Follow the lessons online	
12	Lab: Practice at home by the tutorial: Led array effect.	Online	CLO06		Tutorial Learning KIT		Follow the practice guide	
13	Practice at home Evaluation Lab: - Buttton - Digital Read	Offline	CLO06, CLO09		Tutorial Learning KIT		Discuss with your instructor the lessons learned online and the new lesson.	
14	Practice at home Evaluation Lab: - Input Pullup - Serial - State Change Detection for push buttons	Offline	CLO06, CLO09		Tutorial Learning KIT		Discuss with your instructor the lessons learned online and the new lesson.	
15	Lab: - Analog In, Out Serial - Photoresistor - Analog Input Project Topic Resgistration	Offline	CLO05, CLO06, CLO08, CLO09		Tutorial Learning KIT		Discuss with your instructor the lessons learned online and the new lesson.	
16	Securing IoT	Online	CLO04		Slide Video		Enroll course: https://learning.edx.org/course/course-v1:CurtinX+IOT1x+2T2018/home Follow the lessons online	
17	Securing IoT (cnt)	Online	CLO04		Slide Video		Enroll course: https://learning.edx.org/course/course-v1:CurtinX+IOT1x+2T2018/home Follow the lessons online	
18	Lab: - Practice at home by the tutorial: Photoresistor - States of lights	Online	CLO05, CLO06, CLO09		Tutorial Learning KIT		Follow the practice guide	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
19	Lab: - Photoresistor - while loop interation	Offline	CLO05, CLO06, CLO08, CLO09		Tutorial Learning KIT		Discuss with your instructor the lessons learned online and the new lesson.	
20	Lab: - Ping Ultrasonic Range Finder	Offline	CLO05, CLO06, CLO08, CLO09		Tutorial Learning KIT		Discuss with your instructor the lessons learned online and the new lesson.	
21	Lab: - Servo Motor - Sweep - Servo Motor - Knob	Offline	CLO06, CLO07, CLO08		Tutorial Learning KIT		Discuss with your instructor the lessons learned online and the new lesson.	
22	Electrical circuits and electronics	Online	CLO05		Slide Video		Enroll course: https://learning.edx.org/course/course-v1:CurtinX+IOT2x+2T2018/home Follow the lessons online	
23	Electrical circuits and electronics (cnt)	Online	CLO05		Slide Video		Enroll course: https://learning.edx.org/course/course-v1:CurtinX+IOT2x+2T2018/home Follow the lessons online	
24	Project concept planning	Online	CLO05, CLO10		Slide Video		Enroll course: https://learning.edx.org/course/course-v1:CurtinX+IOT2x+2T2018/home Follow the lesson online	
25	Project concept planning (cnt)	Online	CLO05, CLO10		Slide Video		Enroll course: https://learning.edx.org/course/course-v1:CurtinX+IOT2x+2T2018/home Follow the lesson online	
26	Presentation	Offline	CLO01, CLO02, CLO03, CLO04, CLO05, CLO10				Presentation about Final Project Planning	
27	Presentation	Offline	CLO01, CLO02, CLO03, CLO04, CLO05, CLO10				Presentation about Final Project Planning	
28	Practice at home Evaluation Lab: - IR Remote	Offline	CLO06, CLO07, CLO08		Tutorial Learning KIT		Discuss with your instructor the lessons learned online and the new lesson.	
29	Lab: - Practice at home by the tutorial: Digital Read - Out Serial	Online	CLO05, CLO06, CLO09		Tutorial Learning KIT		Follow the practice guide	
30	Lab: - Practice at home by the tutorial: RGB LED Color Mixing (use tinkercad simulator)	Online	CLO06, CLO07, CLO09		Tutorial Learning KIT		Follow the practice guide	
31	Lab: Practice at home: Advanced Labs related to the Final Project	Online	CLO05, CLO06, CLO07, CLO08, CLO09		Tutorial Learning KIT, Advanced components		Follow the Advanced Lab guide	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
32	Practice at home Evaluation Lab: - Led-7-segment-display	Offline	CLO05, CLO06		Tutorial Learning KIT		Discuss with your instructor the lessons learned online and the new lesson.	
33	Practice at home Evaluation Lab: - Temperature sensor LM35	Offline	CLO05, CLO07, CLO08, CLO09		Tutorial Learning KIT		Discuss with your instructor the lessons learned online and the new lesson.	
34	Progress test 1	Offline	CLO05, CLO06, CLO07, CLO08, CLO09					
35	Lab: - Practice at home by the tutorial: Read ASCII String (use RGB led to indicator)	Online	CLO06, CLO07, CLO09		Tutorial Learning KIT		Follow the practice guide	
36	Lab: - Practice at home by the tutorial: Smoothing Graph: Send data to the computer and graph it in Processing.	Online	CLO07, CLO09, CLO12		Tutorial Learning KIT		Follow the practice guide	
37	Lab: Practice at home: Advanced Labs related to the Final Project	Online	CLO05, CLO06, CLO07, CLO08, CLO09		Tutorial Learning KIT, Advanced components		Follow the Advanced Lab guide	
38	Practice at home Evaluation Lab: - LCD Display 1602	Offline	CLO05, CLO07, CLO08, CLO09		Tutorial Learning KIT		Discuss with your instructor the lessons learned online and the new lesson.	
39	Practice at home Evaluation Lab: - SerialEvent	Offline	CLO13		Tutorial Learning KIT		Discuss with your instructor the lessons learned online and the new lesson.	
40	Practice at home Evaluation Lab: - External Interrupts	Offline	CLO13		Tutorial Learning KIT		Discuss with your instructor the lessons learned online and the new lesson.	
41	Lab: Practice at home by the tutorial: Blink Without Delay	Online	CLO06		Tutorial Learning KIT		Follow the practice guide	
42	Lab: Practice at home by the tutorial: ASCII Table	Online	CLO09		Tutorial Learning KIT		Follow the practice guide	
43	Lab: Practice at home: Advanced Labs related to the Final Project	Online	CLO05, CLO06, CLO07, CLO08, CLO09		Tutorial Learning KIT, Advanced components		Follow the Advanced Lab guide	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
44	Block diagram and Flowchart of Final Project	Online	CLO05, CLO06, CLO07, CLO08, CLO09, CLO10, CLO12				Draw block diagram Create circuit schematic/Hardware inferace/wiring diagram Draw flowchart	
45	Practice at home Evaluation Lab: - Timer Interrupts	Offline	CLO13		Tutorial Learning KIT		Discuss with your instructor the lessons learned online and the new lesson.	
46	On-going project assessment 1	Offline	CLO05, CLO06, CLO07, CLO08, CLO09, CLO10, CLO12				Present Block diagram, Flowchart	
47	On-going project assessment 1 (cnt)	Offline	CLO05, CLO06, CLO07, CLO08, CLO09, CLO10, CLO12				Present Block diagram, Flowchart	
48	Lab: Practice at home by the tutorial: Serial Input	Online	CLO09		Tutorial Learning KIT		Follow the practice guide	
49	Lab: Practice at home by the tutorial: Debounce	Online	CLO06		Tutorial Learning KIT		Follow the practice guide	
50	Practice at home Evaluation Lab: - Keypad 4×4	Offline	CLO06, CLO09, CLO11		Tutorial Learning KIT		Discuss with your instructor the lessons learned online and the new lesson.	
51	Practice at home Evaluation Lab: - EEPROM read / write	Offline	CLO06, CLO09, CLO11		Tutorial Learning KIT		Discuss with your instructor the lessons learned online and the new lesson.	
52	Progress test 2	Offline	CLO01-13					
53	Lab: Practice at home: Advanced Labs related to the Final Project	Online	CLO05, CLO06, CLO07, CLO08, CLO09		Tutorial Learning KIT, Advanced components		Follow the Advanced Lab guide	
54	Final project report preparation	Online	CLO05, CLO06, CLO07, CLO08, CLO09, CLO12		Latex/Overleaf Tutorial online or Report template and guildeline provided by instructor		Prepare Final Report using Latex/Overleaf or Discuss with your instructor for the guidance	
55	Practice at home Evaluation Lab: - 8×8 LED Matrix	Offline	CLO06, CLO07		Tutorial Learning KIT		Discuss with your instructor the lessons learned online and the new lesson.	
56	On-going project assessment 2	Offline	CLO1, CLO10				Prepare for final project presentation	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
57	On-going project assessment 2 (cnt)	Offline	CLO1, CLO10				Prepare for final project presentation	
58	All labs Self review: Practice at home by the tutorial: Team work prepares for the final project	Online			Tutorial Learning KIT		Follow the practice guide	
59	All labs Self review: (cnt) Practice at home by the tutorial: Team work prepares for the final project	Online			Tutorial Learning KIT		Follow the practice guide	
60	All labs Self review: Practice at home by the tutorial: Team work prepares for the final project	Online			Tutorial Learning KIT		Follow the practice guide	

0 Constructive question(s)

6 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Active learning	on-going	1	10.0%	>0	From start to finish in schedule	CLO11, CLO12, CLO13	Private	N/A	Students ask good questions, hard work, self study online.	by instructor	

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
2	Final Project Presentation	on-going	1	20.0%	4	30'/groups	CLO05, CLO06, CLO07, CLO08, CLO09, CLO10, CLO12	Oral	Committee's comments and questions	All chapters; knowledge and skills	The examination is conducted by the exam board. A minimum of two teachers is required for grading the Final Project. Each team is allocated 30 minutes for the presentation of their project followed by a Q&A session. Evaluation of the final presentation is based on the final report, project demonstration, and Q&A interaction. The grade is determined by averaging the grades given by the two teachers.	Based on Rubrics
3	On-Going Project Assessment	on-going	2	30.0%	>0	From start to finish in schedule.	CLO05, CLO06, CLO07, CLO08, CLO09, CLO10, CLO12	Follow lecturer's proposal	Follow lecturer's proposal	knowledge and skills	by instructor	
4	Presentation	on-going	1	10.0%	>0	Option 1: Group. Option 2 (For Constructivism Approach only): Follow lecturer's proposal	CLO01, CLO05	Option 1: N/A. Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Option 1: N/A. Option 2 (For Constructivism Approach only): Follow lecturer's proposal	IoT idea/theory	by instructor	
5	Progress test (Check Practice at home, Exercises, Quiz)	on-going	2	10.0%	>0	Option 1: 30'. Option 2 (For Constructivism Approach only): Follow lecturer's proposal	CLO01, CLO13	Option 1: Private. Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Option 1: N/A. Option 2 (For Constructivism Approach only): Follow lecturer's proposal	studied chapters knowledge and skills	by instructor	
6	Final exam	Final exam	1	20.0%	4	60'	CLO01, CLO13	Computer gradable	50	All chapters; knowledge and skills	by Exam board	